NON-NORMAL ORBIT QUOTIENTS THAT ARE NOT NORMAL QUOTIENTS

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ABSTRACT. Let G be an infinite transitive graph, $\Gamma \leq \mathrm{Aut}(G)$ a transitive group and $A \leq \Gamma$ a subgroup. Grimmett and Li asked whether A can be non-normal in Γ with a transitive orbit quotient G_A that is not isomorphic to G_N for any normal subgroup N of any transitive $\Gamma' \leq \mathrm{Aut}(G)$, and whether a free action of A matters. We show that for every $m \geq 3$ there is an infinite, connected, locally finite, simple Cayley graph X with a non-normal subgroup B of $\mathrm{Aut}(X)$, acting freely, such that X_B is the complete graph K_m, while no normal subgroup of any transitive subgroup of $\mathrm{Aut}(X)$ has K_m as its orbit quotient. The graph is a graphical regular representation of the free product $F * F$, where F is S_3 for $m = 3$ and the alternating group A_m for $m \geq 4$, given by the theorem of Leemann and de la Salle, and B is the preimage of a point stabiliser under the projection to the first factor. The quotient is finite; the question with an infinite quotient is left open.

1. INTRODUCTION

Let $G = (V, E)$ be a connected, infinite, locally finite, simple graph that is transitive, and let $\Gamma \leq \mathrm{Aut}(G)$ act transitively on V. For a subgroup $A \leq \Gamma$, the *simple quotient graph* G_A has the A-orbits as vertices, two distinct orbits Av and Aw being adjacent when some $v' \in Av$ and $w' \in Aw$ are adjacent in G [1, Sect. 4.2]. When A is normal in Γ, the group Γ permutes the A-orbits and acts on G_A by automorphisms, so G_A is transitive; Grimmett and Li proved strict inequalities for connective constants of such quotients [1, Thm. 4.2], [2]. The quotient G_A is defined for non-normal A as well, and Grimmett and Li ask for an example in which the non-normal case is essentially different from the normal one.

Question 1.1 ([1, Question 4.5])**.** Let Γ be a subgroup of $\mathrm{Aut}(G)$ acting transitively on G. Can there exist a non-normal subgroup A of Γ such that: (i) the quotient graph G_A is transitive, and (ii) there exists no normal subgroup N of some transitively acting Γ' such that G_A is isomorphic to G_N? Might it be relevant to assume that A acts freely on V?

Here Γ' ranges over the subgroups of $\mathrm{Aut}(G)$ acting transitively on V. We answer the question as stated, with a free action.

Theorem 1.2. *For every integer $m \geq 3$ there are a connected, infinite, locally finite, simple, transitive graph X and a subgroup $B \leq \mathrm{Aut}(X)$ with the following properties.*

1. *$\mathrm{Aut}(X)$ acts regularly on the vertices of X; B is not normal in $\mathrm{Aut}(X)$, and B acts freely.*
2. *The quotient graph X_B is isomorphic to the complete graph K_m; in particular it is transitive.*
3. *If $\Gamma' \leq \mathrm{Aut}(X)$ acts transitively and $N \trianglelefteq \Gamma'$, then X_N is not isomorphic to K_m.*

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With $\Gamma = \mathrm{Aut}(X)$ and $A = B$, Theorem 1.2 gives a positive answer to Question 1.1 under the extra condition that A acts freely. Part (c) concerns every transitive Γ', but since $\mathrm{Aut}(X)$ acts regularly, the only such Γ' is $\mathrm{Aut}(X)$ itself (Lemma 2.2).

The quotient X_B is finite. The question does not ask for an infinite quotient, but in the setting of connective constants, where a finite quotient has connective constant 0, the interesting version asks for G_A infinite as well. That version is not settled here, and neither is Question 4.6 of [1], which asks whether $\mu(G_A) < \mu(G)$ whenever A is a non-trivial subgroup of Γ acting freely with G_A transitive.

The construction has two ingredients. The first is the theorem of Leemann and de la Salle [3] that a finitely generated group which is not virtually abelian has a Cayley graph of finite degree whose only automorphisms are the left translations; their proof starts from an arbitrary finite generating set and enlarges it, so prescribed edges can be kept. Once the automorphism group is the group itself, acting regularly, normal quotients of X are Cayley graphs of quotient groups, and their number of vertices is an index of a normal subgroup. The second ingredient is a choice of group in which normal subgroups of small index do not exist while non-normal subgroups of index m do: the free product of two copies of a finite group F with a faithful transitive action on m points and no nontrivial homomorphism to a group of order m.

2. Preliminaries

All graphs are simple and undirected. For a group Λ and a finite symmetric generating set $S \subseteq \Lambda \setminus \{1\}$, the Cayley graph $\mathrm{Cay}(\Lambda, S)$ has vertex set Λ and an edge between g and h when $g^{-1}h \in S$. Left multiplication $\lambda_x \colon g \mapsto xg$ is an automorphism, and $\Lambda_L = \{\lambda_x : x \in \Lambda\}$ acts regularly on Λ. The graph is connected because S generates, locally finite of degree $|S|$, and infinite when Λ is.

Theorem 2.1 (Leemann–de la Salle [3, Thm. 1.1 and its proof, Sect. 5])**.** *Let Λ be a finitely generated group that is not virtually abelian, and let $S_0 \subseteq \Lambda \setminus \{1\}$ be a finite symmetric generating set. Put $S_1 = (S_0 \cup S_0^2 \cup S_0^3) \setminus \{1\}$. There is a finite symmetric generating set S with $S_1 \subseteq S \subseteq \Lambda \setminus \{1\}$ and $|S| \le 2|S_1|(|S_1| + 14)$ such that $\mathrm{Aut}(\mathrm{Cay}(\Lambda, S)) = \Lambda_L$.*

In [3], S is the set $\widetilde{S}$ of their Proposition 3.2 applied to S_1, which gives $S_1 \subseteq S$ and the bound on $|S|$; every automorphism of $\mathrm{Cay}(\Lambda, S)$ then preserves the S_1-colours of edges, and their Theorem 3.1 shows that it is a left translation. Theorem 3.1 needs Λ to be neither abelian nor generalized dicyclic; a generalized dicyclic group has an abelian subgroup of index two, so both cases are virtually abelian.

Lemma 2.2. *Let $X = \mathrm{Cay}(\Lambda, S)$ with $\mathrm{Aut}(X) = \Lambda_L$.*

1. *The only subgroup of $\mathrm{Aut}(X)$ acting transitively on Λ is Λ_L.*
2. *If $N \trianglelefteq \Lambda_L$, then $N = \{\lambda_x : x \in N_0\}$ for a normal subgroup $N_0 \trianglelefteq \Lambda$, and X_N has exactly $[\Lambda : N_0]$ vertices.*
3. *For a subgroup $B_0 \le \Lambda$ and $B = \{\lambda_x : x \in B_0\}$, the B-orbits are the right cosets $B_0 g$, and B acts freely.*

Proof. (a) If $\Gamma' \le \Lambda_L$ is transitive, then for every $g \in \Lambda$ some element of Γ' sends 1 to g; the only element of Λ_L doing so is λ_g. Hence $\Gamma' = \Lambda_L$. (b) The map $x \mapsto \lambda_x$ is an isomorphism $\Lambda \to \Lambda_L$, so normal subgroups correspond. The N-orbit of g is $N_0 g = g N_0$, and the orbits are the cosets of N_0. (c) The B-orbit of g is $\{xg : x \in B_0\} = B_0 g$, and $xg = g$ forces $x = 1$. □

3. The construction

Proposition 3.1. *Let $m \geq 3$. Suppose a finite group F acts faithfully and transitively on a set of m points with nontrivial point stabiliser H, and every homomorphism from F to a group of order m is trivial. Then the conclusions of Theorem 1.2 hold for m.*

Proof. Let $\Lambda = F^{(1)} * F^{(2)}$ be the free product of two copies of F, and let $\pi\colon \Lambda \to F$ be the homomorphism that is the identity on $F^{(1)}$ and trivial on $F^{(2)}$. Put $B_0 = \pi^{-1}(H)$. Since π is onto, $[\Lambda : B_0] = [F : H] = m$.

Λ is not virtually abelian. Let $a = |F| - 1 \geq 2$. A reduced word in the free product alternates between nonidentity elements of the two factors, so there are exactly $2a^j$ reduced words of length $j \geq 1$, and by the normal form theorem for free products they represent distinct elements. With respect to the generating set S_0 below, the ball of radius r therefore has at least $2a^r$ elements, and Λ has exponential growth. A finitely generated virtually abelian group has polynomial growth, so Λ is not virtually abelian.

The graph. Let $S_0 = (F^{(1)} \setminus \{1\}) \cup (F^{(2)} \setminus \{1\})$, a finite symmetric generating set of Λ, and let $S \supseteq S_0$ be given by Theorem 2.1. Put $X = \mathrm{Cay}(\Lambda, S)$ and $B = \{\lambda_x : x \in B_0\}$. Then X is connected, infinite, locally finite, simple and transitive, and $\mathrm{Aut}(X) = \Lambda_L$ acts regularly.

(a) If H were normal in F, then, being the stabiliser of one point and conjugate to the stabiliser of every other point by transitivity, it would fix every point; by faithfulness $H = \{1\}$, a contradiction. So there are $f \in F$ and $h \in H$ with $fhf^{-1} \notin H$. Choose $y \in F^{(1)}$ and $x \in B_0$ with $\pi(y) = f$, $\pi(x) = h$; then $\pi(yxy^{-1}) \notin H$, so $yxy^{-1} \notin B_0$ and B_0 is not normal in Λ. Hence B is not normal in $\Lambda_L = \mathrm{Aut}(X)$. It acts freely by Lemma 2.2(c).

(b) By Lemma 2.2(c) the vertices of X_B are the right cosets $B_0 g$. The map $B_0 g \mapsto H\pi(g)$ onto the right cosets of H in F is well defined and injective, because $H\pi(g) = H\pi(g')$ if and only if $\pi(g'g^{-1}) \in H$, that is, $g'g^{-1} \in B_0$; it is onto because π is. Hence X_B has m vertices, and every vertex contains an element of $F^{(1)}$: the coset $B_0 g$ contains the element $r \in F^{(1)}$ with $\pi(r) = \pi(g)$. Given two distinct vertices, pick representatives $r \neq t$ in $F^{(1)}$. Then $r^{-1}t \in F^{(1)} \setminus \{1\} \subseteq S_0 \subseteq S$, so r and t are adjacent in X, and the two vertices are adjacent in X_B. Thus $X_B \cong K_m$.

(c) Let $\Gamma' \leq \mathrm{Aut}(X)$ be transitive and $N \trianglelefteq \Gamma'$ with $X_N \cong K_m$. By Lemma 2.2(a), $\Gamma' = \Lambda_L$, and by Lemma 2.2(b), N corresponds to $N_0 \trianglelefteq \Lambda$ of index m. The quotient map $q\colon \Lambda \to \Lambda/N_0$ is a homomorphism onto a group of order m. Its restrictions to $F^{(1)}$ and to $F^{(2)}$ are homomorphisms from F to a group of order m, hence trivial, and since the two factors generate Λ, q is trivial. This contradicts $|\Lambda/N_0| = m \geq 3$. □

Proof of Theorem 1.2. Let $F = S_3$ if $m = 3$ and $F = A_m$ if $m \geq 4$, acting naturally on $\{1, \ldots, m\}$. The action is faithful. It is transitive: for S_3 clearly, and for A_m the three-cycle $(i\ j\ k)$ sends i to j. The stabiliser of a point has order 2 for $m = 3$ and $(m-1)!/2 \geq 3$ for $m \geq 4$, so it is nontrivial.

Let Q be a group of order m and $\varphi\colon F \to Q$ a homomorphism. If $m = 3$, a transposition has order 2, so its image has order dividing $\gcd(2, 3) = 1$; the transpositions generate S_3, so φ is trivial. If $m = 4$, a three-cycle has order 3 and its image has order dividing $\gcd(3, 4) = 1$; the three-cycles generate A_4, so φ is trivial. If $m \geq 5$, then A_m is simple, so a nontrivial φ is injective, which is impossible since $|A_m| = m!/2 > m$. Proposition 3.1 applies. □

4. Remarks

Remark 4.1 (Degree). With $a = |F| - 1$, the elements of Λ of word length 1, 2 and 3 in S_0 are the reduced words of those lengths, so $|S_1| = t := 2(a + a^2 + a^3)$, and Theorem 2.1 gives a graph of degree at most $2t(t + 14)$. For $m = 3$, $a = 5$, $t = 310$ and the bound is 200,880. The construction is not explicit: the generating set S comes from the probabilistic argument of [3].

Remark 4.2 (The choice of F). The symmetric group S_m does not serve for even $m \geq 4$. The sign maps of the two factors of $S_m * S_m$ give a homomorphism onto the infinite dihedral group $\langle s, t \mid s^2 = t^2 = 1 \rangle$, which maps onto the dihedral group of order m (the Klein four-group when $m = 4$). Its kernel is a normal subgroup of index m, and the abelianization of F alone does not exclude such subgroups. Alternating groups avoid this.

Remark 4.3 (Small m). For $m \leq 2$ the construction cannot work with a graphical regular representation: a subgroup of Λ of index at most 2 is normal, so its orbit quotient is a normal quotient.

Remark 4.4 (Infinite quotients). The argument rests on the fact that Λ has no normal subgroup of index m; for an infinite quotient one would need a non-normal B_0 of infinite index whose Schreier graph with respect to the large generating set S is transitive and differs from every Cayley graph of a quotient of Λ with respect to the image of S. Since S is not explicit, the methods here say nothing about that version of Question 1.1.

Computations

The file `code/check.py` in [4] (Python 3, standard library, exact permutation arithmetic; its output is `code/check_output.txt`) checks, for $F = S_3$ and $F = A_m$ with $4 \leq m \leq 7$: the order of F, transitivity, that the point stabiliser has index m, is nontrivial and is not normal, that the right cosets of the stabiliser are m in number, and, by listing all normal subgroups of F, that none of index greater than 1 has index dividing m. It also checks the count $2a^j$ of reduced words for small a and j and the number 200,880. The proofs above do not depend on this file, and the infinite graph X is not computed.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In September 2026 OpenAI GPT models, running as research tasks of that project, proposed the construction for $m = 3$ and its extension to every $m \geq 3$. In October 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, checked the proof against the sources, wrote the computer checks of the repository [4], and wrote the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code is available at
`https://github.com/yspennstate/nonnormal-orbit-quotients`.

References

[1] G. R. Grimmett and Z. Li, *Self-avoiding walks and connective constants*, in: Sojourns in Probability Theory and Statistical Physics – III, Springer Proc. Math. Stat. **300**, Springer, Singapore, 2019, pp. 215–241; arXiv:1704.05884v2.

[2] G. R. Grimmett and Z. Li, *Strict inequalities for connective constants of transitive graphs*, SIAM J. Discrete Math. **28** (2014), 1306–1333; arXiv:1301.3091.

[3] P.-H. Leemann and M. de la Salle, *Cayley graphs with few automorphisms: the case of infinite groups*, Ann. H. Lebesgue **5** (2022), 73–92; arXiv:2010.06020.

[4] Y. Shmalo, *Code for this paper*, `https://github.com/yspennstate/nonnormal-orbit-quotients`.

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